Sheet 11

Environmental overlap and rates

1. The controlled-environment overlap is $c = 0.3{+}0.4i$, and the input qubit is $+x$. Compute the reduced density matrix, its Bloch vector, purity and $+x$ probability. Can an unread local test tell whether this state arose from an entangled pure composite or a proper preparation mixture?
2. With $\Lambda_{\mathrm{dec}} = 10^{12}\,\mathrm{m^{-2}s^{-1}}$, evaluate the Gaussian kernel multiplier across $d = 1\,\mu\mathrm{m}$ after one second and the corresponding e^{-1} time. For an input with finite momentum variance, find the added momentum variance. Explain why a mathematically valid channel does not by itself justify this quadratic rate for a resolved gas collision.
3. Double the diameter of the lecture's air grain to $20\,\mu\mathrm{m}$, keeping the same bath and resolved separation. Estimate the rate and time. For the separate thermal-photon molecule model, keep its coefficient $10^{10}\,\mathrm{m^{-2}s^{-1}}$ and take separation 20 nm. Find its time. If only its radius doubles within the same Rayleigh model, what factor changes its rate at fixed separation?